UNIVERSITY OF LAHORE

Faculty of Information Technology

Course: Artificial Intelligence

Assignment Title: Customer Churn Prediction Using Machine Learning
Dataset: `WA_Fn-UseC_-Telco-Customer-Churn.xlsx`
Submission: Jupyter Notebook (`.ipynb`) + Final Report (`.pdf`)

Objective

The purpose of this assignment is to build a complete **Customer Churn Prediction** machine learning system. You will analyze customer information and train a model to predict whether a customer is likely to leave the telecom company (**Churn = Yes**) or remain with the company (**Churn = No**).

You may use **VS Code, Jupyter Notebook, Google Colab, PyCharm, or another suitable IDE**, but your final code submission **must include an** `.ipynb` **notebook file**.

Tasks

Complete the following machine learning pipeline:

1. **Data Loading and Understanding:** Load the Excel dataset using Pandas. Display the dataset, shape, column names, data types, basic statistics, and identify the target variable `Churn`. Explain in simple words what the important columns represent.

2. **Data Preprocessing:** Check for missing values, duplicate records, incorrect data types, unnecessary columns, and other data quality problems. Perform the required cleaning and clearly explain why each preprocessing step was needed.

3. **Exploratory Data Analysis:** Explore important patterns in the dataset. Use suitable tables and visualizations to understand customer churn, such as churn distribution and the relationship between churn and important customer characteristics.

4. **Feature Engineering and Encoding:** Select useful input features, create or transform features where appropriate, and convert categorical data into numerical form using suitable encoding techniques. Clearly separate the input features **X** from the target **y**.

5. **Training and Testing:** Split the dataset into training and testing sets. Train at least **two suitable classification algorithms**, such as Logistic Regression, Decision Tree, Random Forest, or another justified model.

6. **Model Evaluation:** Test the trained models using appropriate classification metrics. At minimum, report **Accuracy, Precision, Recall, F1-Score, and Confusion Matrix**. Compare the models and explain which model performs better and why.

7. **Application / Prediction:** Use the selected final model to make churn predictions for sample customer data. Show clearly how the model takes customer information as input and produces a **Churn / No Churn** prediction.

Final Report

Submit a properly formatted **PDF report** explaining the complete work from dataset loading to final prediction. The report must contain **screenshots of your own code execution and outputs** for each major step, along with a short explanation of **what was done, why it was done, and what the result means**. Include a final conclusion discussing the best-performing model, important observations from the data, model limitations, and how such a churn prediction system could be used by a telecom company.

Required Submission

Submit **(1) the completed `.ipynb` notebook, (2) the final PDF report with screenshots and explanations, and (3) any supporting files required to execute your notebook.** Your notebook must run correctly from start to finish. Marks will be based on correct implementation, understanding of the ML pipeline, quality of analysis, model evaluation, application/prediction, and clarity of the final report.

Important: Do not submit only generated code or copied output. You should be able to explain every major step, function, preprocessing decision, and model used in your solution.